AP PRECALCULUS · TOPIC 1.12

Transformations of Functions

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 The point $(1, 5)$ lies on the graph of f. What is the image of this point on the graph of $g(x) = f(x - 3) + 2$?

A. $(4, 7)$ **B.** $(-2, 7)$

C. $(4, 3)$ **D.** $(-2, 3)$

A. $x - 3$ shifts right 3: $x = 4$; $+2$ shifts up 2: $y = 7$. B: horizontal sign error (moved left). C: vertical sign error (moved down). D: both sign errors.

2 What is the vertex of the graph of $g(x) = -2(x + 4)^2 + 1$, a transformation of $f(x) = x^2$?

A. $(4, 1)$ **B.** $(-4, -1)$

C. $(-4, 1)$ **D.** $(-8, 1)$

C. $x + 4$ moves the vertex left 4; $+1$ moves it up 1. The dilation and reflection do not move the vertex. A: horizontal sign error. B: applied the reflection to the vertical translation. D: multiplied the shift by the dilation factor.

3 The point $(6, -3)$ lies on the graph of f. What is the image of this point on the graph of $g(x) = f(2x)$?

A. $(12, -3)$ **B.** $(3, -3)$

C. $(6, -6)$ **D.** $(3, 3)$

B. Horizontal dilation by $\frac{1}{|2|}$: $x = \frac{6}{2} = 3$, y unchanged. A: multiplied x by b instead of dividing. C: treated $f(2x)$ as a vertical dilation. D: also reflected over the x-axis.

4 Let $f(x) = \sqrt{x}$ and $g(x) = f(-x)$. What are the domain and range of g?

A. Domain $[0, \infty)$, range $[0, \infty)$

B. Domain $(-\infty, 0]$, range $(-\infty, 0]$

C. Domain $[0, \infty)$, range $(-\infty, 0]$

D. Domain $(-\infty, 0]$, range $[0, \infty)$

D. $f(-x)$ reflects over the y-axis: $-x \geq 0$ gives domain $(-\infty, 0]$; outputs stay ≥ 0. A: assumed the domain does not change. B: also reflected the range. C: used an x-axis reflection instead.

5 The graph of g is the graph of f shifted left 2 units, dilated vertically by a factor of 3, and reflected over the x-axis. Which equation defines g?

A. $g(x) = -3f(x - 2)$ **B.** $g(x) = -3f(x + 2)$

C. $g(x) = -3f(x) + 2$ **D.** $g(x) = -\frac{1}{3}f(x + 2)$

B. Left 2 is $f(x + 2)$; dilation by 3 with reflection is $-3f$. A: horizontal sign error. C: put the shift outside, making it vertical. D: used the reciprocal as the dilation factor.

6 The function f has domain $[-4, 8]$. What is the domain of $g(x) = f\left(\frac{x}{2}\right)$?

A. $[-2, 4]$ **B.** $[-4, 8]$

C. $[-8, 16]$ **D.** $[-8, 8]$

C. Need $-4 \leq \frac{x}{2} \leq 8$, so $-8 \leq x \leq 16$. A: divided the endpoints by 2. B: assumed the domain is unchanged. D: doubled only the left endpoint.

7 The function f has range $[1, 5]$. What is the range of $g(x) = -2f(x) + 3$?

A. $[-10, -2]$ **B.** $[-16, -8]$

C. $[5, 13]$ **D.** $[-7, 1]$

D. $-2(1) + 3 = 1$ and $-2(5) + 3 = -7$, so the range is $[-7, 1]$. A: forgot the $+3$. B: added 3 before multiplying, computing $-2(f + 3)$. C: ignored the negative sign of -2.

8 The point $(4, 2)$ lies on the graph of f. What is the image of this point on the graph of $g(x) = -f\left(\frac{x}{4}\right) + 1$?

A. $(16, -1)$ **B.** $(1, -1)$

C. $(16, 3)$ **D.** $(16, -3)$

A. x: multiply by $\frac{1}{|1/4|} = 4$ to get 16. y: $-2+1 = -1$. B: divided x by 4. C: missed the reflection. D: added 1 before reflecting, computing $-(2+1)$.

AP PRECALCULUS · TOPIC 1.12 · SECTION 2

Transformations of Functions

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A jet of water from a fountain follows the path $H(x) = -0.4(x-3)^2 + 4.5$, where x is the horizontal distance in metres from the nozzle and H is the height in metres above the water. The path is a transformation of $f(x) = x^2$.

(a) Describe, in order, the transformations of $f(x) = x^2$ that give $H(x)$. DOK3 2 pt

Shift right 3, vertical dilation by 0.4, reflection over the x-axis, shift up 4.5.

(b) Find the greatest height of the jet and the horizontal distance at which it occurs. DOK3 2 pt

The vertex is $(3, 4.5)$: greatest height 4.5 m at $x = 3$ m.

(c) Find the horizontal distance at which the jet returns to the water, to three decimal places. DOK3 2 pt

$-0.4(x-3)^2 + 4.5 = 0 \Rightarrow (x-3)^2 = 11.25 \Rightarrow x = 3 + \sqrt{11.25} \approx 6.354$ m (the positive branch).

Question 10 No calculator · 6 points

The function f has domain $[-3, 5]$ and range $[-2, 6]$, and the point $(2, 4)$ lies on its graph. Let $g(x) = 2f(x-1) + 3$ and $h(x) = -f(2x)$.

(a) Find the domain and the range of g. DOK3 2 pt

Domain: $-3 \le x - 1 \le 5 \Rightarrow [-2, 6]$. Range: $2(-2) + 3 = -1$ to $2(6) + 3 = 15$, so $[-1, 15]$.

(b) Find the image of the point $(2, 4)$ on the graph of g. DOK3 2 pt

$x = 2 + 1 = 3$; $y = 2(4) + 3 = 11$. Image $(3, 11)$.

(c) Find the domain and the range of h. DOK3 2 pt

Domain: $-3 \le 2x \le 5 \Rightarrow \left[-\frac{3}{2}, \frac{5}{2}\right]$. Range: -6 to 2, so $[-6, 2]$.

Answer key (Section 1): 1A 2C 3B 4D 5B 6C 7D 8A